Towards a Multimodal Offline Sensing Architecture for a Semi-Autonomous Social Robot for Child-Robot Interaction

María Gaitán-Padilla[1] *IEEE Student Member*, Jay Patel[2], Camilo A. R. Diaz[1] *IEEE Member*, Carlos A. Cifuentes[2] *IEEE Senior Member*, and Marcela Múnera[2]

***Abstract*— Social robots have demonstrated considerable potential to support education for children by promoting engagement and skill development. However, many existing systems still rely on Wizard-of-Oz operation or cloud-based AI, limiting their autonomy, scalability, and deployment in real-world settings. This work presents the design of a multimodal, offline architecture that enables semi-autonomous interaction for a social robot across educational activities. The proposed system integrates heterogeneous sensing modalities, including vision-based perception, tactile sensing for physical human–robot interaction, distance sensing for proxemic-aware behavior adaptation, offline speech-to-text with keyword recognition, and teachers' and children's inputs through a tablet interface. These data streams are fused by an interaction manager that adapts the robot's behaviour according to the selected activity and the user's interaction state. The architecture supports ten co-created activities targeting academic and socio-emotional skills. The sensing modules have been integrated into the robotic platform and preliminarily validated with adult users, while the activities are currently being integrated into the multimodal decision framework. This work presents the proposed ecosystem and the relationship between multimodal interaction data and adaptive robot behavior.**

I. INTRODUCTION

Socially assistive robots (SARs) are increasingly deployed in healthcare, education, and assistive settings, where they engage in close and prolonged interactions with users [1]. Designing adaptable activities that respond to users' needs has been shown to improve acceptance, engagement, and intention to use across different populations, including autistic children and educational communities [2], [3].

To achieve this adaptability, participatory design (PD) has become an effective methodology for developing technologies that reflect users' real needs and preferences [2]. In social robotics, PD has involved autistic children and teachers in co-designing educational robots [3]. Despite these advances, many educational SAR applications continue to focus on one-to-one interventions rather than supporting inclusive activities involving neurotypical and neurodiverse children in mainstream classrooms [4].

Most existing SAR systems rely on the Wizard-of-Oz (WoZ) paradigm, in which a hidden operator controls the robot's behaviour [5]. Although WoZ facilitates the evaluation of the interaction strategies, it limits scalability, and the robot's behaviour may unintentionally depend on the operator's interpretation of the situation [5]. Semi-autonomous approaches partially address these limitations by allowing autonomous execution of predefined activities while maintaining teachers' supervision when necessary [6].

In child–robot interaction, spontaneous behaviours such as movement, speech, physical contact, and varying attention require robots to perceive multiple interaction cues simultaneously [7]. Consequently, robust multimodal sensing is essential for understanding user engagement, adapting robot behaviour, and ensuring safe and natural human–robot interaction (HRI) in real-world environments [8]. Rather than relying on a single sensing modality, adaptive social robots increasingly require the integration of heterogeneous data streams to provide robust and context-aware decision-making [9]. In addition, deploying AI in child-centred environments raises important concerns regarding data privacy, child emotional development, and system robustness [10]. Following recommendations obtained during the PD process, all perception modules in the proposed system operate locally using offline AI models, avoiding cloud-based processing while protecting sensitive user data and improving deployment robustness.

Therefore, this work presents the ongoing development of a multimodal sensing system for a social robot co-designed to support inclusion. The proposed framework integrates offline AI perception modules to adapt the robot's behaviour during ten co-created educational and socio-emotional games. By combining multimodal sensing with a modular interaction architecture, the system aims to provide semi-autonomous activities while preserving teachers' supervision and supporting inclusive interactions between neurotypical and neurodiverse children in mainstream classrooms.

II. PARTICIPATORY DESIGN

The PD process was conducted in collaboration with a UK primary school to identify user requirements and guide the development of the robot's multimodal interaction capabilities. The process comprised three stages. First, an initial co-design session was held with teachers to identify desirable improvements and define the functionalities to be implemented in a pilot study. Second, the robot was evaluated during a single-session pilot with children, from which feedback was collected from students and teachers. Finally, following an internal brainstorming session within

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[1] M. Gaitán-Padilla and C. A. R. Diaz are with the Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil.

[2] J. Patel, C. A. Cifuentes, and M. Munera are with the Bristol Robotics Laboratory, University of the West of England, Bristol, UK. marcela.munera@uwe.ac.uk

the research team, a second co-design session with the school was conducted to define the requirements for a medium-term deployment protocol in the classroom. The multimodal sensing functionalities presented in this work were derived from the needs identified throughout this iterative PD process.

A. Initial Co-design Session

Twelve staff members from a UK primary school participated in semi-structured interviews to identify desirable improvements to the CASTOR social robot for use with Year 2 children (6–7 years old). The session consisted of two parts: (i) a demonstration of the robot and its existing functionalities, and (ii) semi-structured interviews exploring participants' first impressions, perceived classroom applications, usability, potential risks, and suggestions for future improvements. All participants had experience teaching children aged 4–11 years and were familiar with everyday classroom routines and learning requirements; however, none had previous experience using SARs in educational practice.

The resulting analysis included what teachers liked about the robot, what they disliked, and ideas to improve it to be implemented in a single session in their classrooms. From those ideas, the updates to the robot were planned and developed, as well as the activities to be conducted during the pilot, all depicted in Figure 1.

B. Pilot Feedback

A pilot study was conducted with 88 Year 2 children, divided into six groups of approximately 15 participants following the same interaction protocol. Before interacting with the robot, the children completed a simple show-of-hands questionnaire to assess their initial perceptions regarding the robot, ease of use, social acceptance, and excitement. The interaction session comprised four phases: (i) an introductory conversation to build a good relationship through simple engagement questions; (ii) a Simon Says activity led by the robot; (iii) a storytelling activity supported by multimedia content in the tablet, followed by comprehension questions; and (iv) a feedback session, during which the children repeated the perception questionnaire and teachers evaluated the interaction using the UTAUT model. At the end of the session, the robot offered children the opportunity to say goodbye through a hug, with approximately 92% of participants choosing to interact physically with the robot.

Across the pilot, 27 improvement suggestions were collected from children, teachers, and researchers. Twenty-three of these were translated into functional and hardware requirements that directly informed the next iteration of the system, including the design of new educational activities, multimodal sensing capabilities, robot appearance, and interaction behaviours presented in this work.

C. Brainstorming and Medium-Term Protocol Design

Following the pilot study, the research team conducted a literature review on medium- and long-term deployments of social robots in educational settings, analysing intervention protocols, activities, interaction strategies, and evaluation methods. With these findings and the feedback obtained during the pilot, a collaborative brainstorming session was held to define proposals for a longer integration.

A second co-design session was subsequently held with six members of the partner school's staff working with Year 2 children. Based on their classroom experience and observations from the pilot, the researchers and teachers collaboratively defined the requirements for a medium-term deployment protocol that balanced the research objectives with the school's daily routines. This process also guided the selection of the activities and the multimodal and semi-autonomous interaction requirements described in Section III.

Before the medium-term deployment, the complete interaction framework and educational games will undergo a usability evaluation with teachers to assess the proposed activities, graphical interface, and interaction strategies. The results of this evaluation will support the identification of usability issues, technical limitations, and opportunities for refinement before classroom deployment.

III. Activity Design

The co-designed described in Section II generated a medium-term protocol to implement the social robot within the school's existing classroom routine. The intervention is planned around the *Busy Time* period, during which the robot acts as one of several optional activity stations. During the co-design process, both teachers and students highlighted the importance of increasing the robot's autonomy. Teachers expressed the need to reduce their workload while encouraging collaborative learning and socio-emotional development, whereas children requested more natural interactions, including the ability for the robot to understand spoken responses and react appropriately.

The proposed deployment involves the three Year 2 classrooms (approximately 90 students), with the robot available twice per week in each class. Interaction follows a free-choice model, allowing children to decide whether to interact with the robot individually or in groups of up to fifteen participants. This approach aims to encourage spontaneous social interaction while evaluating the robot in a realistic classroom environment.

The system supports ten educational and socio-emotional activities, each requiring a different combination of perception modules depending on the interaction objectives. Figure 1(b) illustrates the relationship between the activities and the multimodal sensing capabilities integrated into the robot. The activities are described as follows:

- **Maths**: Arithmetic challenges using speech recognition and tablet interaction. Difficulty progresses throughout the intervention, including weekly multiplication practice, while gaze and proxemics are monitored as engagement indicators.
- **Guess How I'm Feeling**: Children identify the robot's emotions while the robot estimates their facial expressions and asks about their emotional state to promote emotional awareness.

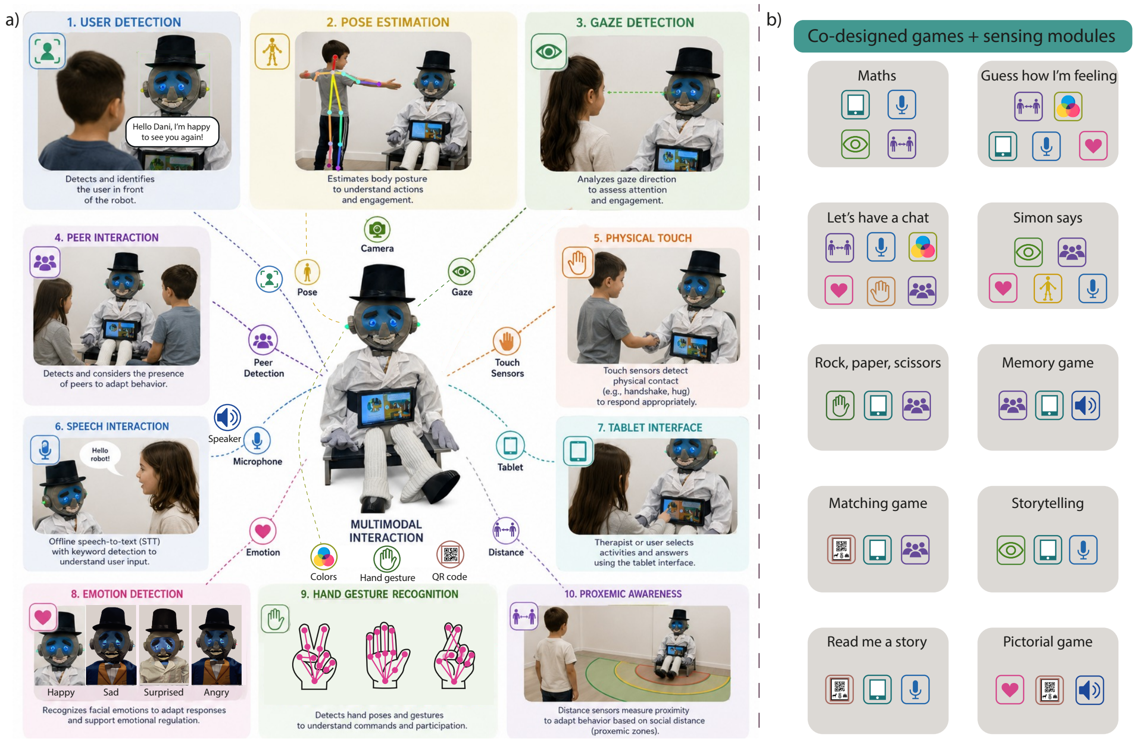

Fig. 1. System architecture. (a) Functions. (b) Activities with the associated modules.

- **Let's Have a Chat**: Guided conversations using offline speech recognition to encourage verbal communication, monitor emotions and peer interaction, and promote physical interactions such as hugs.
- **Simon Says**: A follow-the-instructions game using body pose estimation, emotion recognition, and speech, while monitoring participation and peer interaction.
- **Rock, Paper, Scissors**: A turn-taking game using hand gesture recognition to identify children's choices and display the robot's selection on the tablet.
- **Memory Game**: A collaborative memory game using tablet interaction and verbal positive reinforcement with academic-themed cards.
- **Matching Game**: An educational matching activity using QR-code detection and tablet interaction to associate objects with related concepts.
- **Storytelling**: Interactive storytelling with gaze and emotion monitoring, followed by comprehension questions and positive feedback.
- **Read Me a Story**: Children read aloud while the robot provides encouragement based on keyword detection and engagement cues; communication cards are supported for non-verbal children.
- **Pictogram Game**: A communication activity based on the Picture Exchange Communication System (PECS), allowing children to communicate through categorized pictograms.

IV. System Architecture

The proposed multimodal interaction architecture is illustrated in Figure 1(a). The architecture was designed to enable semi-autonomous HRI by combining multiple offline sensing modalities within a modular perception and decision-making framework. Rather than relying on a single interaction modality, the system integrates heterogeneous data streams to provide context-aware robot behaviours while preserving user privacy through fully offline processing. The interaction begins when one or more children decide to engage with the robot during one of the educational activities. The multimodal sensing layer acquires complementary information through the embedded AI camera, microphone, touch sensors, distance sensors, and tablet interface. Depending on the selected activity, each sensing modality contributes different interaction cues, allowing the system to capture contextual information from the ongoing interaction.

The acquired data are processed by the AI perception layer, which estimates user identity, emotional state, gaze direction, body pose, hand gestures, user position, speech keywords, and QR-code recognition. These modules transform sensor measurements into high-level interaction information that can be shared across different activities. The outputs are then integrated by the interaction manager. This module combines the estimated user state and the current activity state. By maintaining awareness of both the interaction context and the educational activity, the interaction manager coordinates the robot's behaviour while allowing teacher supervision whenever required.

Finally, the adaptive behaviour layer executes the selected responses, including speech reproduction, arms and head movements, color feedback, emotions, and game progression with the tablet interface. Since all activities share the same modular perception pipeline, only the interaction logic changes according to the selected game, enabling the reuse of sensing modules across multiple educational and socio-emotional activities. This modular software architecture facilitates future integration of additional perception capabilities while reducing dependence on WoZ control.

V. Current Development Status

Completed Developments: The participatory design process, including the co-creation of the educational activities and the medium-term deployment protocol, has been completed and provided the functional requirements for the multimodal sensing architecture and interaction framework described in Section III.

The robot has been progressively upgraded by integrating and validating each sensing modality independently. To support physical human–robot interaction (pHRI), two touch-sensing technologies were developed and evaluated to detect physical contact as an engagement indicator and interaction input [11], [12], [13]. For visual perception, previous work explored RGB-D cameras for gaze and face estimation [14]; however, to maintain a low-cost, portable, and fully embedded platform, the system now integrates a HuskyLens2 AI camera[1], providing offline user identification, emotion recognition, gaze estimation, body pose estimation, hand gesture recognition, and user localization [15]. These perception modules enable natural head and eye movements while providing contextual information for activity adaptation.

To enable proxemic-aware behaviour, several distance estimation approaches were investigated, including time-of-flight sensors, camera-based estimation, and redundant sensing through sensor fusion [16], [17]. In addition, an offline speech-to-text pipeline based on Vosk[2] has been incorporated to preserve user privacy while having different languages (English, Portuguese, and Spanish are used). Speech recognition relies on activity-specific keyword detection to trigger appropriate robot responses.

Finally, the robot interface has been redesigned to support WoZ operation and semi-autonomous deployment. A tablet-based graphical interface allows teachers to configure users and activities, while children can independently select and interact with the available educational games.

Ongoing Developments: The ten co-created educational activities are currently being integrated into the multimodal interaction framework and are being evaluated through functional testing with adult volunteers. The modalities are being connected to the specific activity, and after testing them individually, they are being tested together to define the operation and control line in every interaction case. The games are being evaluated through iterative interactions with volunteer students at the university to assess possibilities of errors before continuing with the next game and declaring it ready for the usability test.

[1] https://www.dfrobot.com/product-2995.html
[2] https://github.com/alphacep/vosk-api

VI. Conclusion

This work presents the PD process conducted with teachers and children for the implementation of a SAR in classroom activities. It describes the co-design process, the multimodal semi-autonomous system architecture, the design of ten educational activities with their associated interaction modules, and the planned next stages of development.

As future work, the completed system will undergo a usability evaluation with teachers, followed by a medium-term deployment in collaboration with the partner primary school. The robot will be available during one academic term for approximately 90 Year 2 students to use in free-choice classroom activities.

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